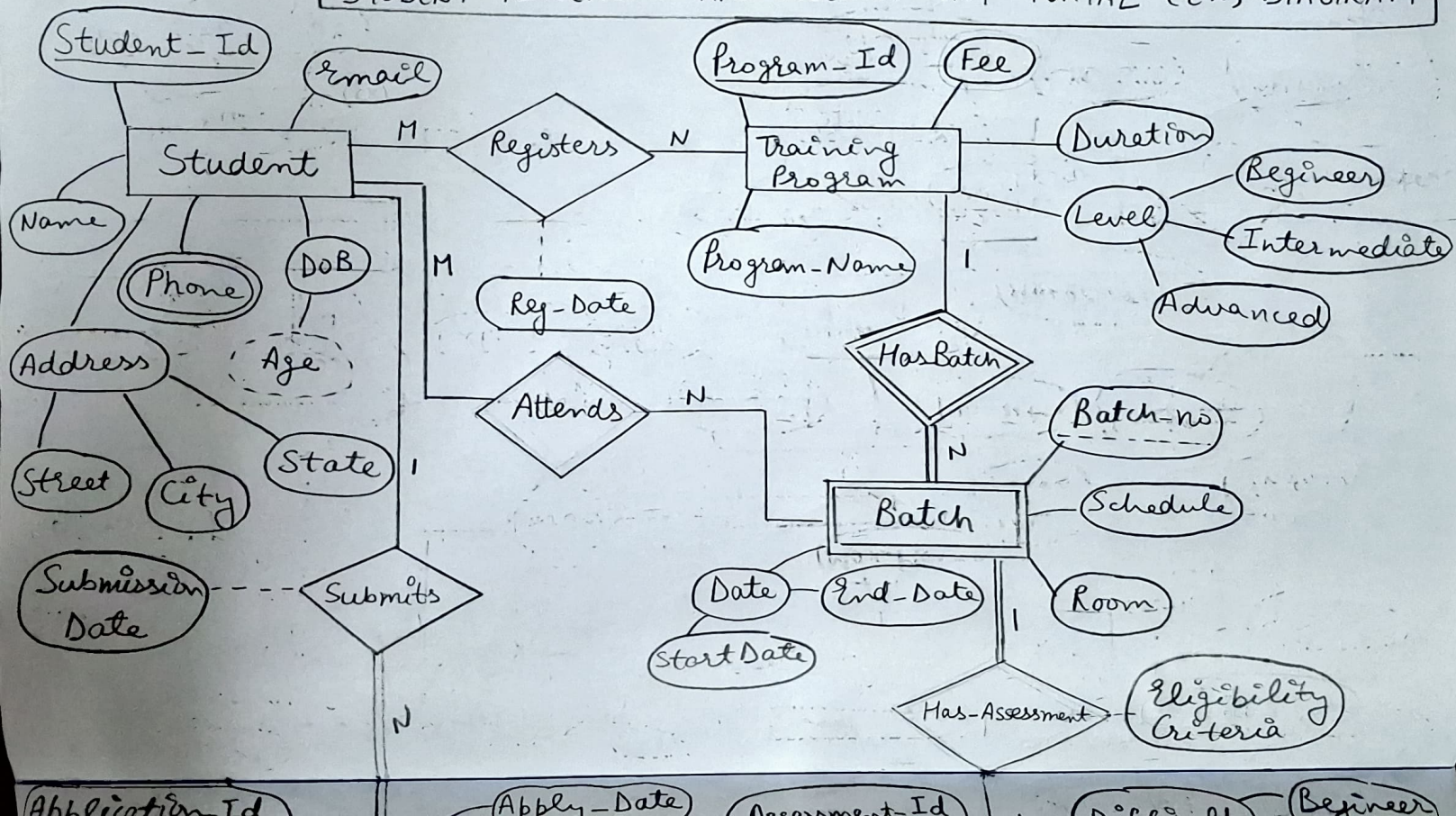
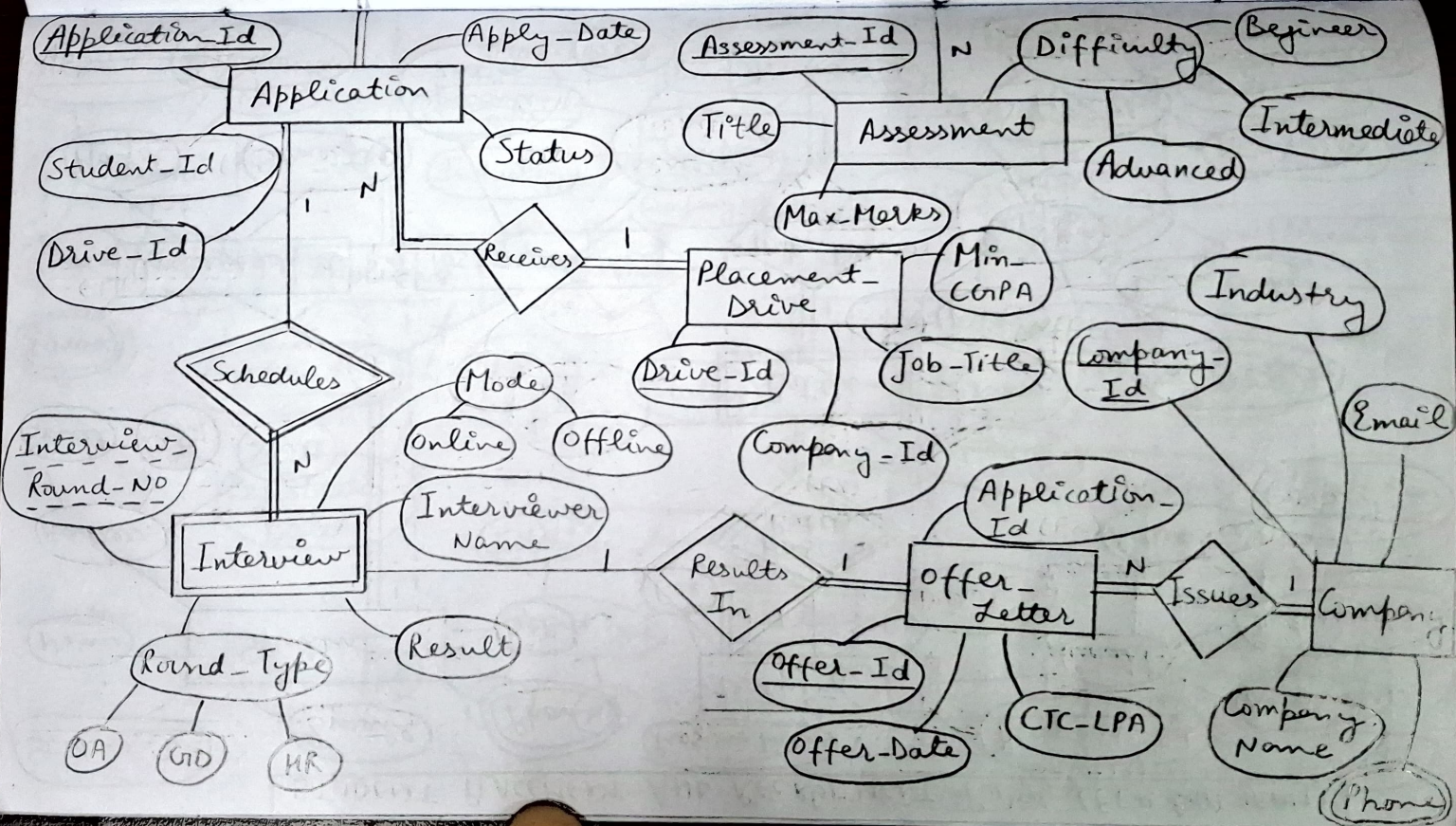
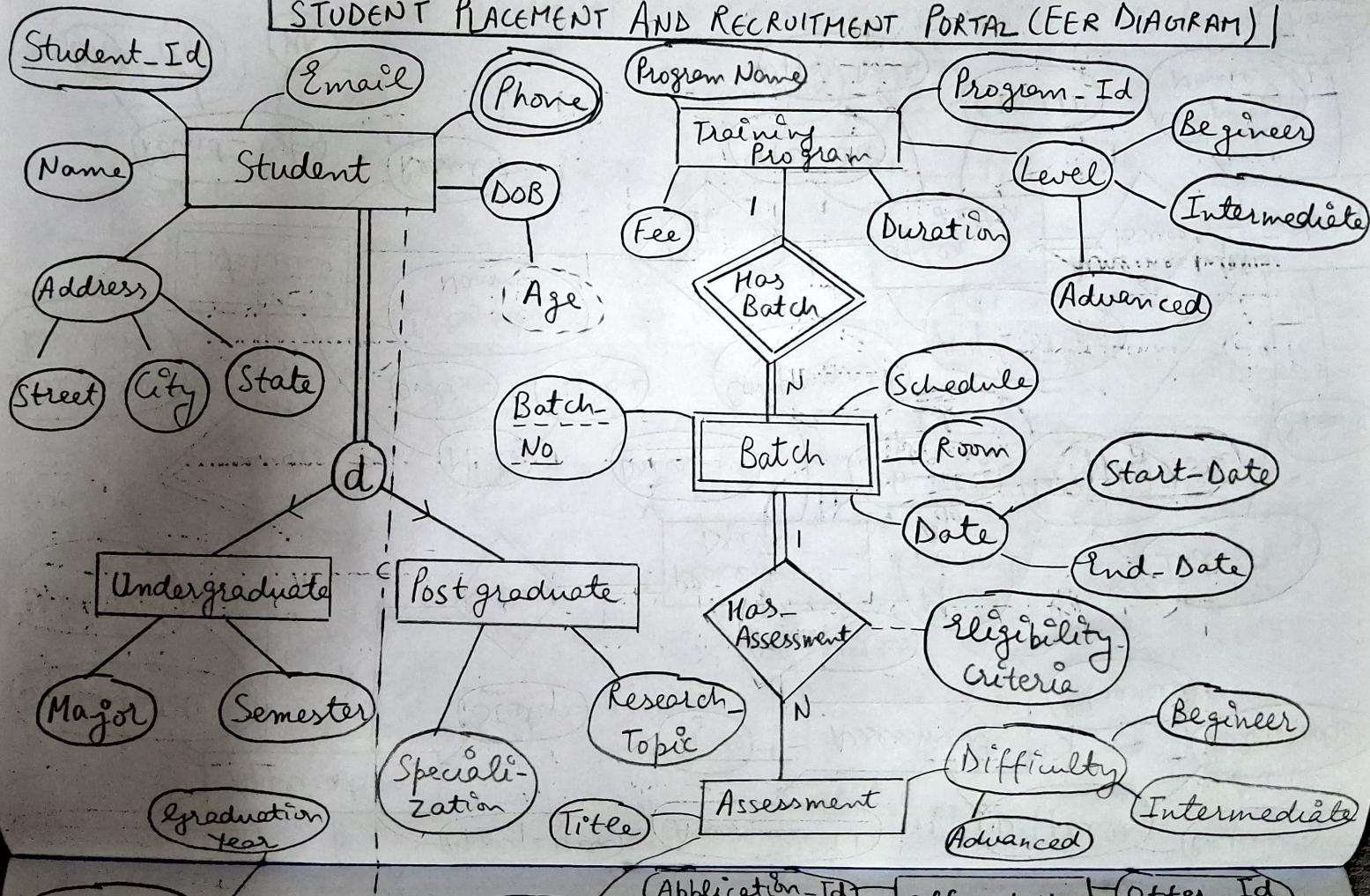


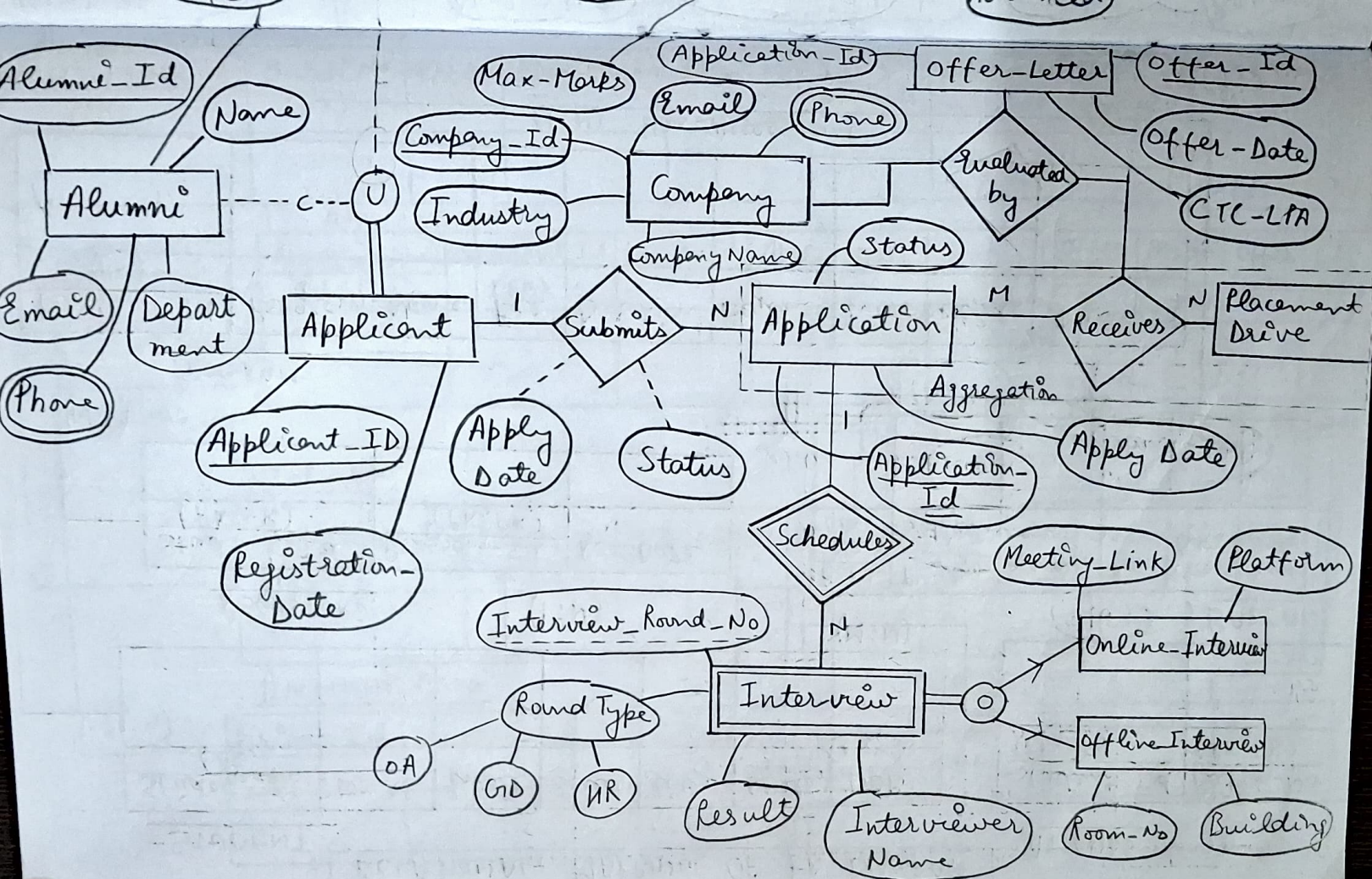
STUDENT PLACEMENT AND RECRUITMENT PORTAL (ER) DIAGRAM





STUDENT PLACEMENT AND RECRUITMENT PORTAL (EER DIAGRAM)





RELATIONAL MAPPING OF ER DIAGRAM

STUDENT

<u>Student-Id</u> (PK)	Name	Email	Street	City	State	DOB

ATTENDS

<u>Student-Id</u> (PK,FK)	<u>Batch-No</u> (PK,FK)

REGISTERS

<u>Student-Id</u> (PK,FK)	<u>Program-Id</u> (PK,FK)	Reg-Date

(1:N)

(M:N)

(1:N)

BATCH

<u>Batch-No</u> (PK)	<u>Program-Id</u> (FK)	Schedule	Room	Start Date	End Date

(N:1)

PROGRAM

<u>Program-Id</u> (PK)	Program-Name	Fee	Duration	Beginner	Intermediate	Advanced

(1:N) Has-Assessment

ASSESSMENT

Assessment-Id	Batch-No	Title	Max-Marks	Beginner	Inter-mediate	Advanced	Eligibil-ity

ASSESSMENT

<u>Assessment-Id</u> (PK)	<u>Batch No</u> (FK)	Title	Max Marks	Beginner	Intermediate	Advanced	Eligibility Criteria

OFFER LETTER

APPLICATION

(1:N) Submits	<u>Application-Id</u> (PK)	<u>Student-Id</u> (FK)	<u>Drive-Id</u> (FK)	Apply-Date	Status	<u>Offer-Id</u> (PK)	<u>Application-Id</u> (FK)	Offer-Date	CTC-LPA
(1:N) Schedules									
Receives									

(1:1) (Results-In)

(N:1) (Issues)

INTERVIEW

<u>Application-Id</u> (PK, FK)	<u>Interview-Round</u> No (PK)	OA	GD	HR	Result	Interviewer Name	Online	Offline

PLACEMENT-DRIVE

(1:N)

<u>Drive-Id</u> (PK)	<u>Company-Id</u> (FK)	Job-Title	Min COPA

COMPANY

<u>Company-Id</u> (PK)	Company-Name	Industry	Email

STUDENT (Multivalued attributes)

<u>Student-Id</u>	<u>Phone (PK)</u>
(PK,FK)	

STUDENT-PHONE

COMPANY (Multivalued attributes)

<u>Company-Id</u>	<u>Phone</u>
(PK,FK)	(PK)

COMPANY-PHONE

TABLE 1 - STUDENT -

<u>Student-Id</u> P(K)	Name	Email	Street	City	State	DOB
ST01	Rahul Sharma	rahul@gmail.com	MG Road	Delhi	Delhi	12-03-2006
ST02	Priya Singh	priya@gmail.com	M1 Road	Jaipur	Rajasthan	05-07-2003
ST03	Amit Kumar	amit@gmail.com	Station Road	Mumbai	Maharashtra	20-01-2006
ST04	Ananya Patel	ananya@gmail.com	FC Road	Pune	Maharashtra	18-11-2005
ST05	Arjun Rao	arjun@gmail.com	Anna Street	Chennai	Tamil Nadu	22-09-2005

Functional Dependency:

Student-Id $\rightarrow$ Name, Email, Street, City, State, DOB

$(\text{Student-Id})^+ = \{ \text{Student-Id, Name, Email, Street, City, State, DOB} \}$

Candidate Key:- Student-Id

1NF:- All attributes contain atomic values. There are no repeating groups or multivalued attributes.

Therefore: STUDENT table is in 1NF.

2NF:- The candidate key is only Student-Id, which is a single attribute. Therefore, partial dependency cannot occur. $\therefore$ STUDENT is in 2NF.

3NF:- No transitive dependency. No dependency exists between non-key (non-prime) attributes.

$\therefore$ STUDENT is in 3NF.

BCNF:- For BCNF, every determinant must be a super/candidate key.

The only non-trivial FD is -
 (Student-Id $\rightarrow$ Name, Email, Street, City, State, DOB)
 Student-Id is a candidate key.
 Therefore, STUDENT is in BCNF.

FINAL SCHEMA -
 STUDENT(
 Student-Id (PK),
 Name,
 Email,
 Street,
 City,
 State,
 DOB
);

[Age not included as it is derived attribute]

TABLE 2 - PROGRAM -

Program-Id (PK)	Program-Name	Fee	Duration	Beginner	Intermediate	Advanced
PRO1	Java Development	12000	12 weeks	Yes	No	No
PRO2	Web Development	15000	16 weeks	No	Yes	No
PRO3	Data Science	18000	20 weeks	No	No	Yes
PRO4	Cloud Computing	14000	14 weeks	No	Yes	No
PRO5	Cyber Security	16000	18 weeks	No	No	Yes

Functional Dependency:

Program-Id $\rightarrow$ Program-Name, Fee, Duration, Beginner, Intermediate, Advanced.

(Program-Id)⁺ = { Program-Id, Program-Name, Fee, Duration, Beginner, Intermediate, Advanced }

Candidate Key:- Program-Id

1NF:- All attributes contain atomic values. There are no repeating groups or multivalued attributes.

Therefore PROGRAM table is in 1NF.

2NF:- The candidate key is only Program-Id, which is a single-attribute. Therefore, partial dependency cannot occur. $\therefore$ PROGRAM is in 2NF.

3NF:- No transitive dependency, no non key (non prime) attribute determines another non key (non prime) attribute. $\therefore$ PROGRAM is in 3NF.

BCNF:- For BCNF, the determinant must be a super/candidate key.

Program-Id is a determinant and is a candidate key.

Therefore, PROGRAM is in BCNF.

FINAL SCHEMA-

PROGRAM(

Program-Id (PK),

Program-Name,

Fee,

Duration,

Beginner,

Intermediate,

Advanced

);

TABLE 3 - REGISTERS -

<u>Student-Id</u> (PK, FK)	<u>Program-Id</u> (PK, FK)	Reg-Date
ST01	PR01	10-01-2026
ST01	PR02	12-01-2026
ST02	PR01	13-01-2026
ST03	PR03	15-01-2026
ST04	PR02	17-01-2026

Functional Dependency :-

$(\text{Student-Id}, \text{Program-Id}) \rightarrow \text{Reg-Date}$

$(\text{Student-Id}, \text{Program-Id})^+ = \{ \text{Student-Id}, \text{Program-Id}, \text{Reg-Date} \}$

Candidate Key - $(\text{Student-Id}, \text{Program-Id})$.

1NF :- All attributes contain atomic values. There are no repeating groups or multivalued attributes.
Therefore REGISTERS is in 1NF.

2NF :- Because the key is composite we must check for any partial dependency.

Neither $\text{Student-Id} \rightarrow \text{Reg-Date}$ nor $\text{Program-Id} \rightarrow \text{Reg-Date}$ holds.

Reg-Date depends on the complete composite key.

Therefore, REGISTERS is in 2NF.

3NF :- There is only one non-key (non-prime) attribute, Reg-Date.

Therefore, no transitive dependency exists.

$\therefore$ REGISTERS is in 3NF.

BCNF: - The determinant. (Student-Id, Program-Id) is a candidate key. Therefore REGISTERS is in BCNF.

FINAL SCHEMA -

REGISTERS (

Student-Id PK, FK,

Program-Id PK, FK,

Reg-Date

);

TABLE 4 - BATCH -

PROGRAM-ID	PROGRAM-NAME	BATCH-NO	BATCH-DETAILS	START DATE	END-DATE
PRO1	Java Dev	B01	Morning, R101	01/02/26	30/04/26
PRO1	Java Dev	B02	Evening, R102	05/02/26	05/05/26
PRO2	web Dev	B01	Morning, R201	10/02/26	10/06/26

INF Conversion -

Program-Id	Program-Name	Batch-No	Schedule	start Date	end Date	Room
PRO1	Java Dev	B01	Morning	01/02/26	30/04/26	R101
PRO1	Java Dev	B02	Evening	05/02/26	05/05/26	R102
PRO2	web Dev	B01	Morning	10/02/26	10/06/26	R201

Functional Dependency:-

- (Program-ID, Batch-No) → Schedule, Room, Start Date, End Date
- Program-ID → Program-Name (Partial Dependency)

{Program

Candidate

2NF DEC

Because
of the
a Part

Program -

PRO1

PRO1

PRO2

PRO3

PRO4

3NF

Room

R101

R102

R201

attribute

(no trans

The t

BCNF -

Schedule

$\{ \text{Program-Id, Batch-No} \}^+$ $\left\{ \begin{array}{l} \text{Program-Id, } \cancel{\text{Program-Name}} \\ \text{Batch-No, Schedule,} \\ \text{Room, start Date, End Date,} \end{array} \right\}$

Candidate Key:- (Program-Id, Batch-No)

Program-Name $\uparrow$

2NF DECOMPOSITION -

Because Program-Name depends on only one part of the candidate key (Program-Id), it creates a Partial Dependency (a 2NF violation).

Program-Id	Batch-No	Schedule	Room	Start-Date	End-Date
PRO1	B01	Morning	R101	01/02/26	30/04/26
PRO1	B02	Evening	R102	05/02/26	05/05/26
PRO2	B01	Morning	R201	10/02/26	10/06/26
PRO3	B01	Evening	R301	15/02/26	15/07/26
PRO4	B01	Morning	R401	20/02/26	20/05/26

3NF (NO PARTIAL TRANSITIVE DEPENDENCY) -

The remaining FD in the table is -

$FD_1: (\text{Program-Id, Batch-No}) \rightarrow \text{Schedule, Room, Start-Date, End-Date.}$

(Program-Id, Batch-No) is the Candidate

Key (Super Key), there are no non-key attributes determining other non-key attributes (no transitive dependency like $A \rightarrow B$ and $B \rightarrow C$)

The table is already in 3NF.

BCNF - $(FD_1): (\text{Program-Id, Batch-No}) \rightarrow \text{Schedule, Room, Start-Date, End-Date}$

(Program-Id, Batch-no) is a valid candidate key. Therefore, the relation/table is in BCNF.

FINAL SCHEMA -

BATCH (

Program-Id (PK/FK),

Batch-no (PK),

Schedule,

Room,

Start-Date,

End-Date

);

TABLE 5 - ATTENDS -

<u>Student-Id</u> (FK)	Program-Id	<u>Batch-No</u> (FK)
ST01	PR01	B01
ST01	PR02	B01
ST02	PR01	B02
ST04	PR02	B02
ST05	PR03	B01

Functional Dependencies -

FD₁: (Student-Id, Program-Id) → Batch-No

FD₂: Batch-No → Program-Id (if Batch-Id's are globally unique to program across institute).

Candidate Key -

CK₁ = (Student-Id, Program-Id)

CK₂ = (Student-Id, Batch-No)

(Student-Id, Program-Id) ^{T_s} { Student-Id,
Program-Id, Batch-No }

$(\text{Student-Id}, \text{Batch-No})^+ = \left\{ \begin{matrix} \text{Student-Id}, \text{Batch-No} \\ \text{No}, \text{Program Id} \end{matrix} \right\}$

1NF - All attributes contain atomic values (no multivalued attributes).

Therefore ATTENDS is in 1NF.

2NF - All attributes are prime attributes. There are no non-prime attributes to violate 2NF. Therefore, the table is in 2NF.

3NF - $(FD)_1: (\text{Student-Id}, \text{Program-Id}) \rightarrow \text{Batch-No}$
 $(\text{Student-Id}, \text{Program-Id})$ is a CK/SK.
 Batch-No is a prime attribute.

$(FD)_2: \text{Batch-No} \rightarrow \text{Program-Id}$.
 Batch-No is not a super key on its own.
 But Program-Id is a prime attribute.

Therefore, ATTENDS table is in 3NF.

BCNF - Violation in $FD_2: \text{Batch-No} \rightarrow \text{Program-Id}$,
 Batch-No is not a super key. Fails BCNF.

Decomposition: Separate into Batch-Program and Attends.

Batch-Program

Batch-No	Program-Id
B01	PR01
B02	PR01
B01	PR02
B02	PR02
B01	PR03

Attends

Student-Id	Batch-No (FK)
ST01	B01
ST02	B02
ST03	B01
ST04	B02
ST05	B01

FINAL SCHEMA -

Batch-Program (

Batch-No (PK),

Program-Id

);

Attends (

student-Id (PK)

Batch-No (FK)

);

TABLE 6 - ASSESSMENTS -

Assessment-Id	Program-Id	Batch-No	Title	Max-Marks	Level-Flags	Eligibility-Criteria
AS01	PRO1	B01	Java Basics	100	Beginner, Level-1	60% Attendance
AS02	PRO1	B02	OOP Test	100	Intermediate, Level-2	70% Attendance
AS03	PRO2	B01	Web Dev Quiz	50	Beginner, Level-1	65% Attendance
AS04	PRO2	B02	React Test	100	Intermediate, Level-2	75% Attendance
AS05	PRO3	B01	Data Struts	100	Advanced, Level-3	80% Attendance

INF CONVERSION - split the non-atomic level-Flags into atomic boolean columns (Beginner, Intermediate, Advanced).

Assessment-Id	Program-Id	Batch-No	Title	Max-Marks	Beginner	Intermediate	Advanced	Eligibility-Criteria
AS01	PRO1	B01	Java Basics	100	Yes	No	No	60% Attendance
AS02	PRO1	B02	OOP Test	100	No	Yes	No	70% Attendance
AS03	PRO2	B01	Web Dev Quiz	50	Yes	No	No	65% Attendance
AS04	PRO2	B02	React-Test	100	No	Yes	No	75% Attendance
AS05	PRO3	B01	Data Struts	100	No	No	Yes	80% Attendance

- Candidate Keys: $CK_1 = (\text{Assessment-ID})$,
 $CK_2 = (\text{Batch-No}, \text{Title}, \text{Program-ID})$

- Functional Dependencies -

$\rightarrow FD_1: \text{Assessment-ID} \rightarrow \text{Program-ID}, \text{Batch-No},$
 $\text{Title}, \text{Max-Marks}, \text{Beginner}, \text{Intermediate},$
 $\text{Advanced}, \text{Eligibility-Criteria}$

$\rightarrow FD_2: (\text{Batch-No}, \text{Title}) \rightarrow \text{Program-ID}$ (
 Batch and Assessment Title uniquely
 identify the program).

2NF - Both ~~LHS~~ for FD_1 and FD_2 , the
 LHS is candidate key and RHS is a prime
 attribute.

CK_1 is a single attribute, no partial
 dependency exists.

For CK_2 RHS is prime attribute, no partial
 dependency.

3NF -

- For FD_1 : Determinant $X = \text{Assessment-ID}$
 is a superkey.
- FD_2 : Determinant $(\text{Batch-No}, \text{Title})$ is NOT a
 superkey, but $Y = \text{Program-ID}$ is a prime
 attribute in CK_2 .

$\therefore$ The table is in 3NF.

BCNF -

violation in $FD_2: (\text{Batch-No}, \text{Title}) \rightarrow$
 Program-ID is not a superkey. Fails BCNF

- Decomposition: split into BATCH-TITLE-PROGRAM
 and ASSESSMENT.

BATCH - TITLE - PROGRAM

<u>Batch-no (PK)</u>	<u>Title (PK)</u>	<u>Program - Id</u>
B01	Java Basics	PR01
B02	oop Test	PR01
B01	web Dev Quiz	PR02
B02	React Test	PR02
B01	Data Structs	PR03

ASSESSMENT

<u>Assessment - Id</u> (PK)	<u>Batch - No</u> (FK)	<u>Title</u> (FK)	<u>Max</u> <u>Marks</u>	<u>Beginner</u>	<u>Intermediate</u>	<u>Advanced</u>	<u>Eligibility</u> <u>Criteria</u>
AS01	B01	Java Basics	100	Yes	No	No	60% Attendance
AS02	B02	oop Test	100	No	Yes	No	70% Attendance
AS03	B01	web Dev Quiz	50	Yes	No	No	65% Attendance
AS04	B02	React Test	100	No	Yes	No	75% Attendance
AS05	B01	Data Structs	100	No	No	Yes	80% Attendance

TABLE - 7 - COMPANY -

<u>Company - Id (PK)</u>	<u>Company - Name</u>	<u>Industry</u>	<u>Email</u>
C01	Google	Tech	careers@gmail.com
C02	Microsoft	Tech	careers@ms.com
C03	Amazon	E-commerce	jobs@amazon.com
C04	Meta	Social Media	talent@meta.com
C05	Apple	Hardware	hr@apple.com

1NF - The given table contains all atomic attribute values and has no multivalued attributes. Therefore, the table is in 1NF.

• Candidate Keys:-

$CK_1 = (\text{Company-Id})$, $CK_2 = (\text{Company-Name, Industry, Email})$

• Functional Dependencies:-

$FD_1: \text{Company-Id} \rightarrow \text{Company-Name, Industry, Email}$

$FD_2: \text{Email} \rightarrow \text{Company-Name}$ (Company corporate email domain determines Company Name)

2NF - $FD_1: \text{Company-Id}$ is a single-attribute (PK). Non-prime attributes depend directly on Company-Id. There is no partial dependency.

FD_2 : Therefore, the table is in 2NF.

3NF - For $FD_1: \text{Company-Id} \rightarrow \text{Company-Name, Industry, Email}$.

Company-Id is a valid super key.

For $FD_2: \text{Email} \rightarrow \text{Company-Name}$

Email is not a superkey.

$\{\text{Email}\}^+ \subset \{\text{Email, Company-Name}\}$

and Company-Name is non-prime attribute because the only CK is Company-Id.

DECOMPOSITION - To eliminate transitive dependency

$FD_2: \text{Email} \rightarrow \text{Company-Name}$, decompose the table into:

EMAIL - COMPANY : (Email (PK), Company - Name)

COMPANY: (Company - Id (PK), Email (FK), Industry)

EMAIL - COMPANY

Email (PK)	Company - Name
careers@google.com	Google
careers@ms.com	Microsoft
jobs@amazon.com	Amazon
talent@meta.com	Meta
hr@apple.com	Apple

COMPANY

Company - Id (PK)	Email (FK)	Industry
Co1	careers@google.com	Tech.
Co2	careers@ms.com	Tech
Co3	jobs@amazon.com	E-commerce
Co4	talent@meta.com	Social Media
Co5	hr@apple.com	Hardware

BCNF - The two tables are in BCNF because
For both the FD_1 and FD_2 .

FD_1 : Company - Id is a valid (CK/SK) of
COMPANY table.

FD_2 : Email is a valid (CK/SK) of EMAIL -
COMPANY table

FINAL SCHEMA-

EMAIL - COMPANY (
 email (PK),
 Company - Name
);

COMPANY (
 Company - Id (PK),
 email (FK),
 Industry
);

TABLE - 8 - PLACEMENT_DRIVE -

<u>Drive - Id</u>	<u>Company - Id</u>	<u>Job - Info</u>	<u>Min CRPA</u>
D01	Co1	Software Engineer, Full time	8.0
D02	Co2	Cloud Engineer, Full Time	7.5
D03	Co3	System Analyst, Full Time	7.0
D04	Co4	AI Research Engineer, Full Time	8.5
D05	Co5	Hardware Engineer, Full Time	7.8

INF - Extract the atomic Job - Title attribute from Job - Info.

<u>Drive - Id</u>	<u>Company - Id</u>	<u>Job - Title</u>	<u>Min CRPA</u>
D01	Co1	Software Engineer	8.0
D02	Co2	Cloud Engineer	7.5
D03	Co3	System Analyst	7.0
D04	Co4	AI Research Engineer	8.5
D05	Co5	Hardware Engineer	7.8

Candidate Keys: $CK_1 = (\text{Drive-Id})$. ~~also~~ $CK_2 = (\text{Company-Id})$

• Functional Dependencies: -

$FD_1: \text{Drive-Id} \rightarrow \text{Company-Id}, \text{Job-Title}, \text{Min-CGPA}$

$FD_2: \text{Job-Title} \rightarrow \text{Company-Id}$ (each unique Job-Title is mapped exclusively to one target Company.)

2NF - Drive-Id is a single-attribute CK.
Therefore, no partial dependency can exist.
 $\therefore$ the table PLACEMENT_DRIVE is in 2NF.

3NF -

$FD_1: \text{Drive-Id} \rightarrow \text{Company-Id}, \text{Job-Title}, \text{Min-CGPA}$

Drive-Id is a candidate key / super key.

$FD_2: \text{Job-Title} \rightarrow \text{Company-Id}$

• Job-Title is not a super key.

• Company-Id is non-prime because it is not a part any CK.

$\therefore FD_2$ violates 3NF.

DECOMPOSITION - To eliminate transitive dependency

$FD_2: \text{Job-Title} \rightarrow \text{Company-Id}$, decompose the table into:-

JOB-COMPANY : (Job-Title (PK), Company-Id).

PLACEMENT_DRIVE : (Drive-Id (PK), Job-Title (FK), Min-CGPA)

JOB - COMPANY -

<u>Job - Title (PK)</u>	<u>Company - Id</u>
Software Engineer	Co1
Cloud Engineer	Co2
Systems Analyst	Co3
AI Research Engineer	Co4
Hardware Engineer	Co5

PLACEMENT - DRIVE -

<u>Drive - Id (PK)</u>	<u>Job - Title (FK)</u>	<u>Min - CorPA</u>
D01	Software Engineer	8.0
D02	Cloud Engineer	7.5
D03	Systems Analyst	7.0
D04	AI Research Engineer	8.5
D05	Hardware Engineer	7.8

BCNF - Since, for both the tables, JOB - COMPANY and PLACEMENT - DRIVE LHS is a Super Key, therefore both of them satisfies conditions for BCNF.

∴ JOB - COMPANY and PLACEMENT - DRIVE both are in BCNF.

FINAL SCHEMA -
 JOB - COMPANY (
 Job - Title (PK),
 Company - Id
);

PLACEMENT - DRIVE (
 Drive - Id (PK),
 Job - Title (FK),
 Min CGPA
);

TABLE - 9 - APPLICATION -

<u>Application - Id</u>	Student - Id	Drive Id	Apply - Date	Status.
A01	ST01	D01	10/03/26	Shortlisted
A02	ST01	D02	12/03/26	Applied
A03	ST02	D03	10/03/26	Rejected
A04	ST03	D04	11/03/26	Interviewing
A05	ST04	D05	15/03/26	Shortlisted

1NF - The given table contains all atomic attributes (values) and there are no multi valued attributes. Therefore, the table is in 1NF.

Candidate key : (Application - Id)

Functional - Dependencies -

- FD_1 : Application - Id $\rightarrow$ Student - Id, Drive - Id, Apply - Date, Status.
- FD_2 : (Student - Id, Apply - Date) $\rightarrow$ Drive - Id.

2NF - ~~Both~~ candidate keys: Application-Id is a single-attribute candidate key.

Therefore, partial dependency cannot exist.

Therefore, the table is in 2NF.

3NF - For FD₁: Application-Id $\rightarrow$ Student-Id, Drive-Id, Apply-Date, Status.

(Application-Id is a candidate key.)

For FD₂: (Student-Id, Apply-Date) $\rightarrow$ Drive-Id

- (Student-Id, Apply-Date) is not a superkey
- Drive-Id is non-prime because it is not a part of candidate key.

$\therefore$ FD₂ violates the conditions for 3NF.

DECOMPOSITION - To eliminate the transitive dependency, decompose the table into.

STUDENT_DAILY-ROUTINE-DRIVE : (Student-Id(PK), Apply-Date(PK), Drive-Id)

APPLICATION : (Application-Id(PK), Student-Id(FK), Apply-Date(FK), Status).

STUDENT_DAILY-ROUTINE-DRIVE-

<u>Student-Id</u> (PK)	<u>Apply-Date</u> (PK)	Drive-Id
ST01	10/03/26	D01
ST01	12/03/26	D02
ST02	10/03/26	D03
ST03	11/03/26	D04
ST04	15/03/26	D05

APPLICATION -

Application - Id (PK)	Student - Id (FK)	Apply - Date (FK)	Status
A01	ST01	10/03/26	Shortlisted
A02	ST01	12/03/26	Applied
A03	ST02	10/03/26	Rejected
A04	ST03	11/03/26	Interviewing
A05	ST04	15/03/26	Shortlisted

BCNF - since, for both the FDs, FD_1 and FD_2 (Application - Id) is a super key for FD_1 and (Student - Id, Apply - Date) is a super key for FD_2 ∴ the given tables are in BCNF.

FINAL SCHEMA -

STUDENT-DAILY-ROUTING-DRIVE (

Student - Id (PK),

Apply - Date (PK),

Drive - Id

);

APPLICATION (

Application - Id (PK),

Student - Id (FK),

Apply - Date (FK),

Status.

);

TABLE - 10 - INTERVIEW -

Application-Id	Interview-Round-No	OA	GD	HR	Result
A01	1	Yes	No	No	Pass
A02	2	No	No	Yes	Pass
A03	1	Yes	No	No	Pass
A04	2	No	Yes	No	Pending
A05	3	No	No	Yes	Pass

1NF - The given table contains all atomic values and there are no multivalued attributes present. Therefore, the table is in 1NF.

Candidate Keys -

$CK_1 = (\text{Application-Id}, \text{Interview-Round-No})$

$CK_2 = (\text{Application-Id}, \text{Interviewer-Name})$

Functional Dependencies -

• $FD_1: (\text{Application-Id}, \text{Interview-Round-No}) \rightarrow \text{OA}, \text{GD}, \text{HR}, \text{Result}, \text{Interviewer-Name}, \text{online}, \text{offline}$

• $FD_2: \text{Interviewer-Name} \rightarrow \text{Interview-Round-No}$
(Each Interviewer is designated strictly to conduct a specific interview round)

2NF - Both Candidate keys are composite

There are no dependencies where a non-prime attribute depends on only a proper subset of either candidate key. (In 2NF)

Interviewer-Name	online	offline
Raj-Mehta	Yes	No
Neha Singh	No	Yes
Vikram Patel	Yes	No
Ananya Roy	Yes	No
Suresh Kumar	No	Yes

3NF - $\nexists$ For FD_1 : the LHS (Application-Id, Interview-Round-No) is a CK.

$\therefore$ It is in 3NF.

For $FD_2: (\text{Interviewer-Name}) \rightarrow (\text{Interview-Round-No})$

LHS is not a super key,

but RHS is a prime attribute because it is a part of CK.

Therefore, FD_2 satisfies the condition for 3NF.

$\therefore$ It is in 3NF.

BCNF - For FD_1 the LHS is a super key.

$\therefore$ It is in BCNF.

But, for FD_2 the LHS is not a super key,

violates the condition for BCNF.

DECOMPOSITION - To ~~decompose~~^{convert}, the table into BCNF decompose it into.

INTERVIEWER-ROUNDS: (Interviewer-Name(PK),

Interviewer - Round - No)

INTERVIEW : (Application - Id (PK), Interviewer - Name (PK|FK), OA, GD, HR, Result, online, offline).

INTERVIEWER - ROUND -

<u>Interviewer - Name (PK)</u>	<u>Interview - Round No</u>
Raj Mehta	1
Neha Singh	2
Vikram Patel	1
Ananya Roy	2
Suresh Kumar	3

INTERVIEW -

<u>Application - Id (PK)</u>	<u>Interviewer Name (PK FK)</u>	OA	GD	HR	Result	online	offline
A01	Raj Mehta	Yes	No	No	Pass	Yes	No
A01	Neha Singh	No	No	Yes	Pass	No	Yes
A02	Vikram Patel	Yes	No	No	Pass	Yes	No
A04	Ananya Roy	No	Yes	Yes	Pending	Yes	No
A05	Suresh Kumar	No	No	Yes	Pass	No	Yes

Both the tables are now in BCNF.

FINAL SCHEMA -

INTERVIEWER - ROUND (
Interviewer - Name (PK),
Interview - Round - No
);

INTERVIEW (

Application - Id (PK),
Interviewer Name (PK|FK),
OA,
GD,
HR,
online, offline);

TABLE-11 - OFFER-LETTER -

offer-Id	Application-Id	offer-Details
001	A01	Date: 25/03/26, CTC: 12.5 LPA
002	A02	Date: 27/03/26, CTC: 10.0 LPA
003	A03	Date: 28/03/26, CTC: 8.5 LPA
004	A04	Date: 29/03/26, CTC: 15.0 LPA
005	A05	Date: 30/03/26, CTC: 11.0 LPA

INF - split offer-Details into atomic offer-Details and CTC-LPA.

offer-Id	Application-Id	offer-Date	CTC-LPA
001	A01	25/03/26	12.5
002	A02	27/03/26	10.0
003	A03	28/03/26	8.5
004	A04	29/03/26	15.0
005	A05	30/03/26	11.0

Candidate Keys: - $(CK_1) = (\text{offer-Id})$,
 $CK_2 = (\text{Application-Id})$

Functional Dependencies:

FD1: offer-Id $\rightarrow$ Application-Id, offer-Date, CTC-LPA

FD2: Application-Id $\rightarrow$ offer-Id, offer-Date, CTC-LPA
 Id (each application can receive at most one offer).

2NF - Both candidate keys are single attributes.
Therefore, partial dependencies cannot exist.

$\therefore$ the table is in 2NF.

3NF - For -

FD₁: Offer - Id $\rightarrow$ Application - Id, Offer - Date,
CTC - LPA

(Offer - Id) is a candidate key.

FD₂: Application - Id $\rightarrow$ Offer - Id, Offer - Date,
CTC - LPA

Application - Id is also a candidate key.

$\therefore$ Both the FDs satisfies 3NF.

$\therefore$ The table is in 3NF.

BCNF - Every determinant is a candidate key.

Offer - Id $\rightarrow$ candidate key.

Application - Id $\rightarrow$ candidate key.

$\therefore$ the given table is in BCNF.

FINAL SCHEMA -

offer LETTER (

offer - Id (PK),

Application - Id,

offer - Date,

CTC - LPA

);

TABLE - 12 - STUDENT - PHONE -

<u>Student - Id</u>	<u>Student - Name</u>	<u>Phone</u>
ST01	Rahul Sharma	9867543210,
ST02	Riya Singh	9123456789
ST03	Amit Patel	8675432109
ST04	Sneha Reddy	7654321089, 9987
ST05	Vikram Kumar	6543210789
		9569267241

1NF - Separate the multi-valued attribute into atomic records

STUDENT - PHONE

<u>Student - Id</u>	<u>Phone</u>
ST01	9867543210
ST01	9123456789
ST02	8675432109
ST03	7654321089
ST03	9987654321
ST04	6543210789
ST05	9569267241

Candidate keys: (Student - Id, Phone - No)

2NF - Both attributes form the composite candidate key. There are no non-prime attributes, so no partial dependency exists. $\therefore$ the table is in 2NF.

3NF - There are no non-trivial FDs involving non-key attributes. $\therefore$ the table is in 3NF.

BCNF - The only determinant is the CK.
Therefore every determinant is a super key.

∴ the table is in BCNF.

FINAL SCHEMA -

STUDENT-PHONE (

Student-Id (PK),

Phone (PK)

);

TABLE - 13 - COMPANY - PHONE

<u>Company - Id</u>	<u>Company - Name</u>	<u>Phone</u>
C01	Google	18004190123,
C02	Microsoft	18004190124
C03	Amazon	18001020001
C04	Meta	18003000111, 18003000112
C05	Apple	18004000123
		18005000111,
		18005000112

INF - Since, Phone contains multiple values relation is not in INF.

<u>Company - Id</u>	<u>Phone</u>
C01	18004190123
C01	18004190124
C02	18001020001
C03	18003000111
C03	18003000112
C04	18004000123
C05	18005000111
C05	18005000112

Candidate Key :- (Company - Id, Phone - No)

2NF - Both attributes are part of composite candidate key.

There are no non-prime attributes.

Therefore, no partial dependency exists.

Results :- Passes 2NF.

3NF - There are no non-trivial FDs involving non-key attributes.

∴ The table is in 3NF.

BCNF - The only determinant is the candidate key (Company - Id, Phone - No).

Therefore, the table is in BCNF.

FINAL SCHEMA -

COMPANY - PHONE

Company - Id (PK),

Phone No (PK)

);